Toward Load Identification Based on the Hilbert Transform and Sequence to Sequence Long Short-Term Memory

Thi-Thu-Huong Le, Shinwook Heo, and Howon Kim, *Member, IEEE*

***Abstract*—Load identification is a core concept in non-intrusive load monitoring (NILM). Through NILM systems, users can check their home appliance usage habits and then adjust their behavior to save electricity. In this way, a NILM system offers an effective method to detect the event status of household appliances as well as individual loads' energy consumption. However, prior NILM methods have encountered a challenge in improving recognition accuracy for both linear load and non-linear load types. These methods used a representative feature, namely transient load signals. However, the transient signals on these loads differ in terms of transient time and transient shape, which is the main cause of reduced accuracy performance in load identification. To this end, this paper presents a novel method, HT-LSTM (Hilbert Transform Long Short-Term Memory), which enhances recognition of the various load types that contain the difference in the transient time and the transient shape of any load signal. The proposed method consists of two main parts: (i) generating a novel transient feature based on a Hilbert transform (HT), called APF (Amplitude-Phase-Frequency). APF features are sequential data, which is used for the classification model; and (ii) applying Sequence-to-Sequence Long Short-Term Memory (Seq2Seq LSTM) to identify appliances by using APF features as the input data. In this work, we evaluate the HT-LSTM method using two high-frequency public datasets, Building-Level fUlly-labeled dataset for Electricity Disaggregation (BLUED) and Plug Load Appliance Identification Dataset (PLAID). Also, we evaluate our method using a private dataset collected in the lab. Based on the experimental results obtained and comparison classification performance pointed, the proposed method outperforms previous methods of F-score measurement on both public datasets in load identification as well as the private dataset.**

***Index Terms*—NILM, smart grid, smart home, transient-state, high-frequency data, frequency domain, Hilbert transform, Seq2Seq LSTM.**

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I. Introduction

RECENTLY, the smart grid has been attracting the attention not only of researchers but also of many national governments. In particular, electrical energy is one of the most important forms of energy and needs effective management to save power. Electricity demand increases substantially every year, with the building sectors consuming the most electricity. According to [1], electricity usage in residential and commercial buildings makes up 74% of total electricity consumption, with residential buildings making up 52% of U.S. electricity usage in 2010. Again, 63% of total EU electricity consumption is accounted for by residential buildings [2]. Thus, energy saving has become an urgent energy goal globally. Hart proposed the first non-intrusive load monitoring (NILM) concept [3]. NILM measures the individual home appliance's usage using only a single sensor to record aggregate power consumption. This innovative feature offers reduced deployment costs. Hence, NILM technology demonstrates the potential next-generation electrical network.

In the last years, machine learning has played a significant role in research on NILM. In the literature, solutions based on Nearest Neighbour (k-NN), support vector machine (SVM), and different variants of hidden Markov Models (HMMs) have been proposed [4]. Especially, the proposed disaggregator based on HMM model and a new Viterbi algorithm by Makonin *et al.* [5] can run in real-time on an inexpensive embedded processing using low-frequency, Reference Energy Disaggregation Data Set (REDD) dataset. However, this method still the exponential limitation in time and space because their disaggregator meets this limitation issue when the number of super-states are more larger [5].

Recently, approaches based on deep learning have received particular attention since they exhibited remarkable load identification performance. Deep Neural Network (DNNs) have been successfully applied for the first time to NILM by Kelly and Knottebelt (2015) in [6]. Next, variant LSTM models have been successfully applied for low-frequency NILM datasets of UK-DALE (UK Domestic Appliance-Level Electricity), REDD, and AMPds (Almanac of Minutely Power dataset) in [7]–[8]. Furthermore, increasing more numerical results have been published using low-frequency datasets since 2019,

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for examples, Transferability of neural networks with CNN (Convolutional Neural Network) and GRU (Gated Recurrent Unit) [9] (2019) using UK-DALE, REDD, and REFIT (Personalised Retrofit Decision Support Tools For UK Homes Using Smart Home Technology) dataset at low sampling rate; Deep Temporal Dictionary Learning (DTDL) method for energy disaggregation (2019) by Khodayar *et al.* [10] using low-frequency REDD dataset; Context Bidirectional LSTM (CoBiLSTM) [11] (2020) using low-frequency data, for example, AMPds, REFIT, and REDD; Power Generative Adversarial Network (PowerGAN) [12] using low-sampling data of REFIT; and so on. In the literature, most of the NILM researchers above focused their works on deep learning methods, especially deep neural networks to solve load identification or energy disaggregation using low-frequency data. The reason is that low-frequency data acquisition enables a low-cost design because only a single sensor is needed. The data obtained is less noisy and no need to transform the domain data to extract the feature data. The feature factors at a low-frequency commonly comprise a root mean square (RMS) current, RMS voltage, active power (P), reactive power (Q), and so on. However, this occurs the imbalance the number of researches using data types of low-frequency data and high-frequency data. Although high-frequency data acquisition involves a high-cost design, including additional hardware as well as a sensor, for example, a current device to prevent electrical white noise, feature factors such as current or voltage at a high sampling rate contain much valuable information, creating unique features for recognizing various load types.

Although almost numerical results are focused on the classification to solve the NILM issue using low-frequency data, some researchers recently have been focusing on providing their successful high-frequency NILM methods. Moreover, deep neural networks combined with frequency processing also are applied to high-frequency NILM datasets to solve load identification or load disaggregation. In particular, the authors [13] proposed FIT-PS (Frequency Invariant Transform of Periodic Signals) hybrids LSTM net using HELD1 (Home Equipment Laboratory Dataset 1) dataset with a sampling frequency of 4 kHz. Their highest classification accuracy result obtained 87.9% for the training process and 90% for the detection rate with 18 different devices. Besides, Wu and Wang [14] proposed Concatenate-DenseNet-121 on image processing obtained by converted time domain to frequency domain based on STFT (Short-Time Fourier Transform), using high-frequency BLUED dataset. The obtained F1 score of this method was 83.9%. Furthermore, the previous research [15] proposed a method called FFT-BDT (Fast Fourier Transform Bagging Decision Tree) to address the challenge of detecting household appliances with similar power consumption using high-frequency data, PLAID. The achieved accuracy and F1 measurements of this research were 92.8% and 92.48%, respectively. Therefore, it is possible to improve load recognition performance for the learning model by using high-frequency data.

In this paper, we aim to make contributions to overcome the current NILM issue for load identification using high-frequency data by proposing a novel supervised method, HT-LSTM. First, generating a unique and effective novel transient feature named APF (Amplitude-Phase-Frequency) based on the Hilbert transform (HT). APF features are created from instantaneous amplitude, instantaneous phase, and instantaneous frequency of the analytic signal by HT. APF can support the sequence learning model to recognize various load types through differences in the transient time and the transient shape of load signals. Then, applying the Sequence-to-Sequence Long Short-Term Memory (Seq2Seq LSTM) model to learn on the novel transient-state feature. The novel NILM method is named HT-LSTM (Hilbert Transform Long Short-Term Memory), to overcome a current NILM challenge. The nature of the challenge is described in detail in Section II. In this case, we perform evaluations of the HT-LSTM method using two high-frequency publicly datasets, BLUED and PLAID, and a private dataset collected at the lab. Also, we compare the performance classification of the proposed method to other previous methods on two similar high-frequency publicly datasets. Based on experiment results and comparison, our method can enhance performance classification of F1 measurement on high sampling rate data of various residential load types.

The remainder of the paper is structured as follows. Section II states the problem, related to previous NILM studies. Section III presents the proposed method in detail. Then, Section IV presents experimental evaluation results and comparison with prior methods. Finally, Section V concludes our paper.

II. Related Work and Problem Statement

A. Related Work

The transient-state feature approach is a standard method for NILM using data obtained at a high sampling rate [16]. The data can be used to create unique features for improving the performance of the learning model. In digital signal processing, the transient-state feature can be extracted by the transforming technique. The raw current and voltage data are waveforms formed in the time domain. To obtain richer analyzing information of raw current or voltage waveforms, these raw data often are transformed from the time domain to the frequency domain using some transform techniques. Several transient features have been created from raw data, for examples, transient response time, electromagnetic interference (EMI), the spectral envelope, and so on. In particular, voltage noise was extracted using a fast Fourier transform (FFT) that can detect SMPS loads. However, this method required noise that is specific to each appliance [17]. Besides, spectral envelopes and transient profiles created by transient power [18]–[19] can be used to detect several types of appliance. Nevertheless, these features require a repeatable transient profile setting.

Furthermore, transient response time which affects discrete wavelet transform (DWT) coupled with a multilayer feed-forward neural network (MFNN) [20]. However, this method performed accurate detection only on linear load types such as vacuum cleaners. Besides, EMI as a unique, time-invariant feature for load identification as per [21]. This feature is useful for SMPS loads, for examples, LED lamps, laptop chargers, speakers, and so on. However, it cannot be used to detect circuit loads that have no SMPS model, for example, refrigerators, blenders, irons, room heaters, televisions,

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and so on. Moreover, a transient feature in the spectrogram, in time-limited to 3 seconds using STFT and CNN model [14] could not detect loads with a transient time of more than 3 seconds, such as a treadmill (running machine) in high-frequency BLUED dataset. Nevertheless, several recent high-frequency NILM methods, for examples, DWT-Decision Tree [22], FFT-BDT [15], and STFT-GAN-CNN [23], have been successfully obtained average classification accuracy over 90%, based on DTW, FFT, and STFT transformation methods respectively combined to decision tree, bagging decision tree, and GAN-CNN models using high-frequency PLAID and LIT (Laboratory for Innovation and Technology) datasets, respectively.

On the other hand, a few researchers proposed their methods using raw current and voltage data without transforming the domain. For examples, Baets *et al.* [24], adopted a CNN with VI trajectories to automatically extract features and thereby classify appliances. However, the F1 measurement of the method does not offer high performance, at 77.60% for the PLAID dataset. Besides, Barsim *et al.* [25] proposed a neural network-based method to identify loads by creating VI-trajectories of selected raw high-resolution current and voltage waveforms from the PLAID dataset with the accuracy obtained 89.7%.

Based on the transient-state feature approach analysis above, our first observation is possible to create a unique transient-state feature of high-frequency data for the chosen learning model. This feature can be used to detect any load type that contains differences in the transient time and shape of the transient load signal.

B. Problem Statement

For alternating current (AC) appliances, the load types are distinguished based on how these loads consume power. Most household appliances fall into one of the two load types, linear and non-linear. Some researchers may be unfamiliar with these characteristics of linear and non-linear loads due to a lack of specialist knowledge of power systems. Hence, we analyze the main characteristics of each load type in this work. In particular, resistive, inductive, and capacitive loads belong to the linear load category, while non-linear loads have an SMPS element. We can categorize a load as either linear or non-linear depending on whether it draws current along with the voltage waveform, varying with a pure sinusoidal pattern. This feature discloses load properties that we can leverage to handle in our learning models.

The first category of the appliance is the linear load type. For a linear load type, the load is capacitive, as opposed to a resistive or inductive load. (1) The capacitive load is built into a circuit so that the system can be tested without being connected to its real-world load. Although there are capacitive elements in loads, inductive and resistive elements still dominate. Besides, the residential building sector is restricted to use household appliances, which feature capacitive loads. Hence, in this work, we focus on analyzing only resistive and inductive loads. From the circuit perspective, these linear loads have a few different features. (2) The resistive load has a heating element as a linear load. The resistive load covers devices containing any type of heating element, for examples, incandescent lights, coffee makers, kettles, hairdryers, ovens, toasters, heaters, and so on. Resistive loads draw current in a sinusoidal waveform that is in phase with the voltage. Hence, the current waveform is either a purely sinusoid signal or a distorted signal. (3) Inductive load contains AC motors elements inside of loads. Vacuum cleaners, washing machines, air-conditioners, fans, etc. are representative examples of inductive loads. Inductive load draws current along the sinusoidal pattern lags the voltage waveform. Hence, the current waveform is either a purely sinusoidal signal or a distorted signal non-linear load that has SMPS. The current waveform is always a distorted signal. Resistive and inductive loads have similar characteristics when switched on. Heating elements and AC motors have stable current values with smooth sinusoidal waveforms, except for certain individual load cases such as a hairdryer or iron (resistive load) or a fan (inductive load).

The second category of the appliance is the non-linear load type, containing an SMPS element. This load type does not draw a sinusoidal current. From the circuit viewpoint, non-linear loads have different features compared to linear loads. Electronic devices, for example, desktop computers and TVs that contain SMPS elements are representative examples of the most predominant type of non-linear load. Another example of a non-linear load type is fluorescent lights. Further, a load that converts AC to a low-voltage direct current (DC) is also non-linear, an example being battery chargers.

In addition, resistive and inductive loads have similar characteristics when switched on. Heating elements and AC motors have stable current values with smooth sinusoidal waveforms, except for certain individual load cases such as a hairdryer or iron (resistive load) or a fan (inductive load). On the other hand, the non-linear loads with SMPS do not draw a sinusoidal current, presenting distorted current waveforms. Therefore, one of the challenges in load identification is the various types of loads and their different characteristics [26]–[27].

Based on various types of load analysis above, our second observation is possible to construct a learning model that can detect various load types effectively, including linear and non-linear loads, using high-frequency data.

From two observations above, a current challenge for researchers in developing a high-frequency NILM method is necessary to enhance performance classification of load identification in recognizing various load type that contains differences in transient time and shape. To address this challenge, we need a proposal for a novel transient feature using energy data with a high sampling rate. This novel feature is generated from raw various load types including inductive load, resistive load, and SMPS load of high-frequency data. This novel feature data is used for the input data of the learning model to improve the recognition of various residential appliances.

III. The Proposed Load Identification Method: HT-LSTM

This section describes the load identification method proposed to address the current challenge, described in Section II, for NILM following the transient-state approach.

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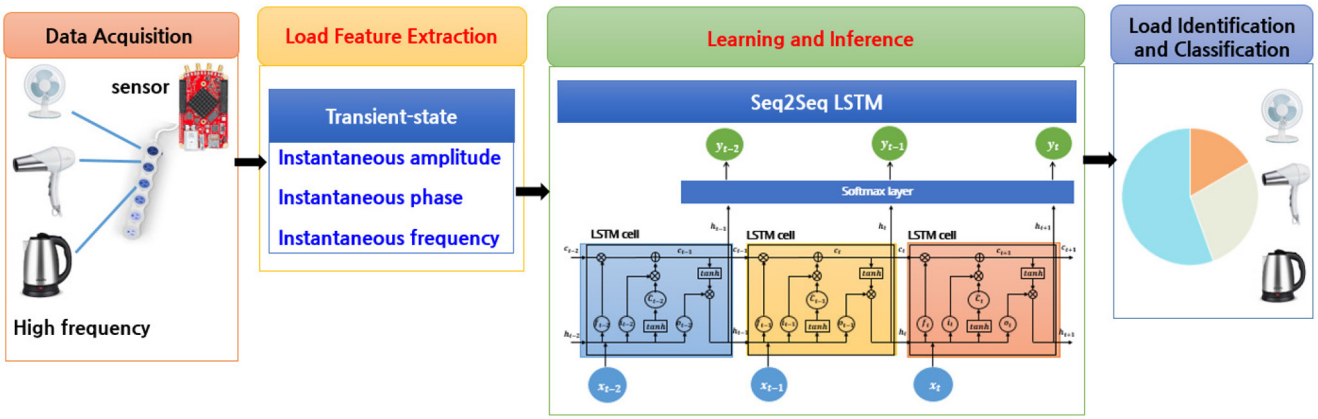


Fig. 1. The architecture of the proposed method HT-LSTM.

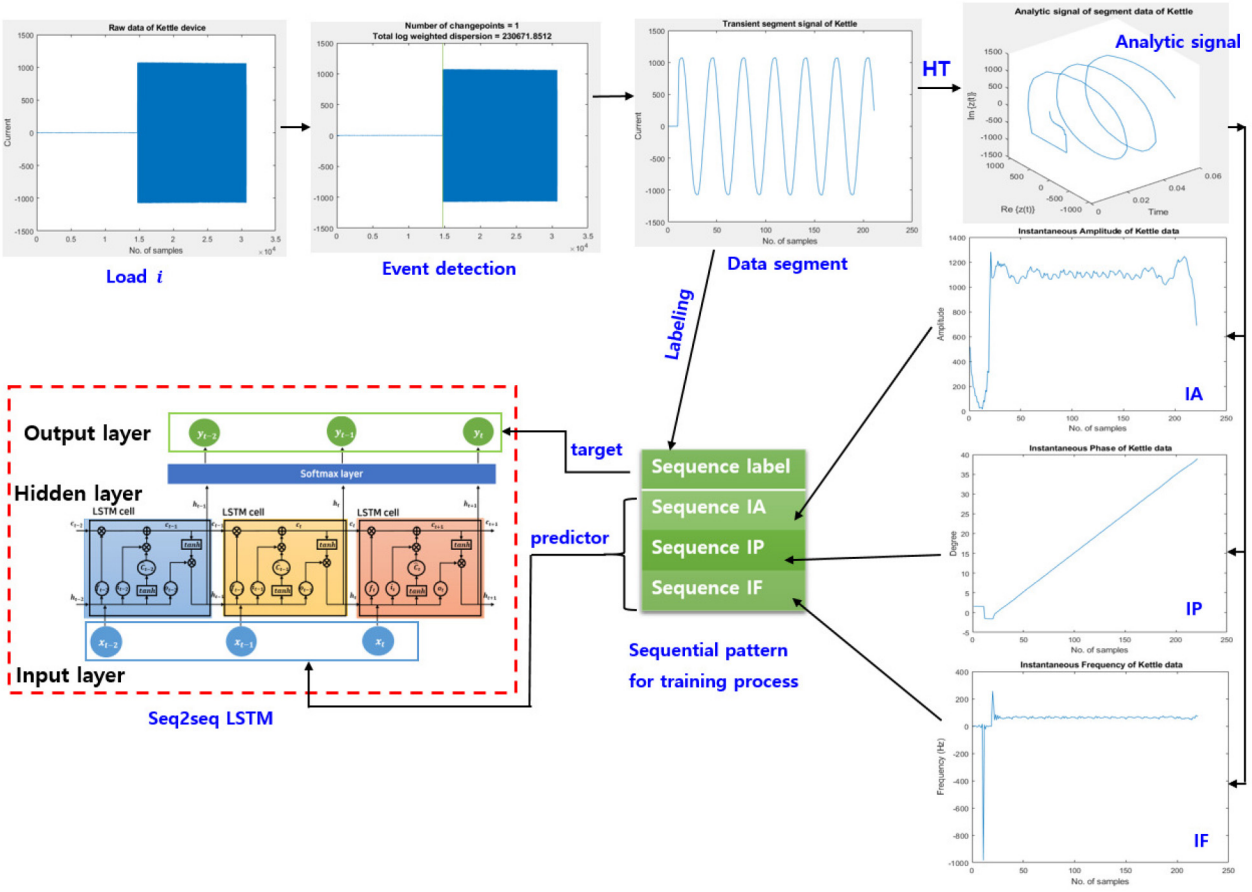


Fig. 2. The HT-LSTM training process.

The method can effectively detect various load types based on differences in transient time and shape. In outline, this section includes an illustration of the concept of the HT-LSTM method, the proposed algorithm for extracting our novel APF feature, and the applied learning model and its algorithm for enhancing the classification of loads. At first, the proposed architecture is shown in Figure 1.

A. Concept Illustration of HT-LSTM

The key idea of HT-LSTM is to build a learning process with the Seq2Seq LSTM model based on the generated transient-state features. Because the APF feature is generated and extracted from transient-state data for the current signal, the APF feature is categorized in the transient-state feature approach. The transient-state features extracted are IA (Instantaneous Amplitude), IP (Instantaneous Phase), and IF (Instantaneous Frequency). More specifically, the proposed method extracts novel IA, IP, and IF features using the original current data. IA, IP, and IF are generated as the input data of the classification model. We then use the trained Seq2Seq LSTM model to identify household appliances on testing data. The HT-LSTM method has two phases:

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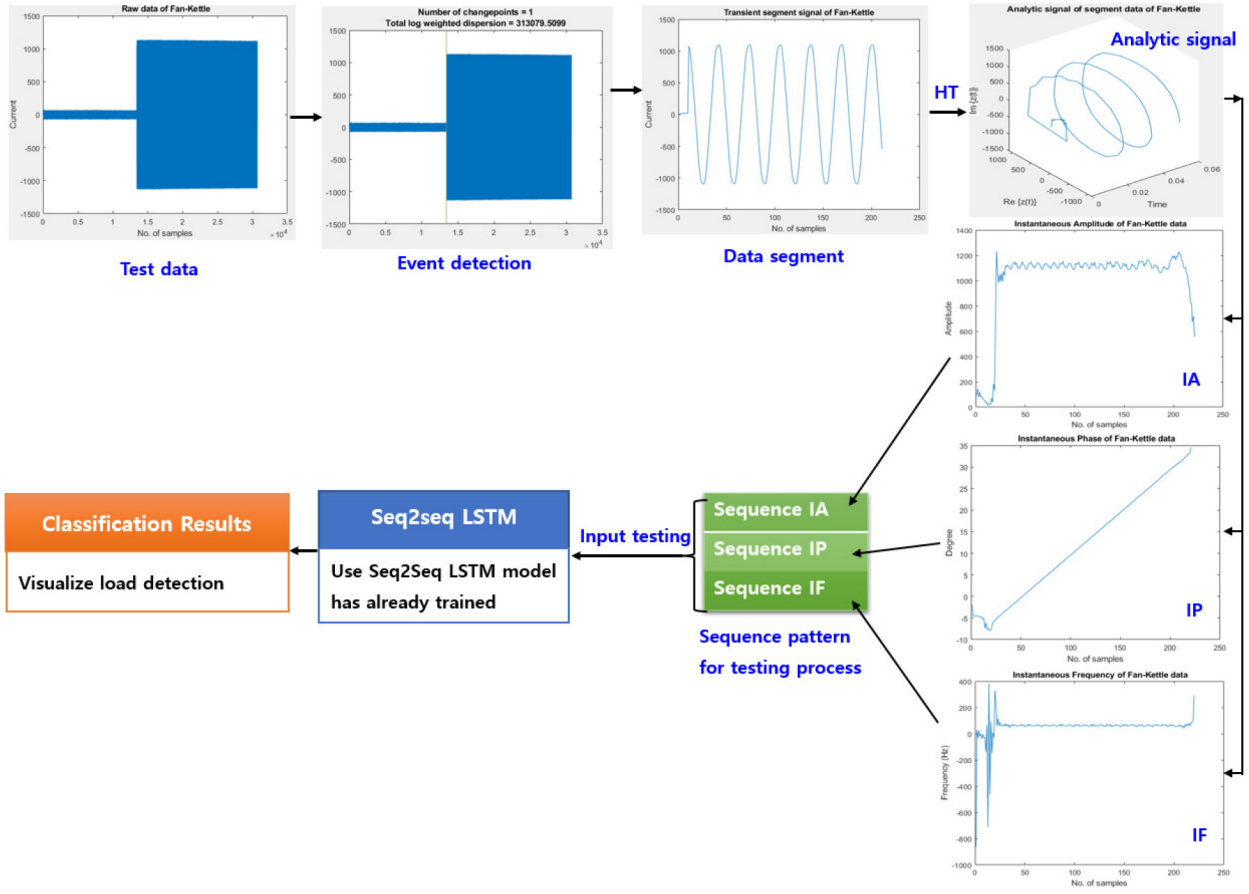


Fig. 3. The HT-LSTM testing process.

(1) Training phase: In this phase, there are several steps. Step 1 is event detection using the RMS threshold. In this step, we could select transient segment data from raw current data using the occurred-event index (when users switch on the device). Step 2 is feature extraction; at this step, we create IA, IP, and IF information by applying the Hilbert Transform (HT) algorithm. IA, IP, and IF values are used to create novel APF transient features. In parallel, we create the sequential labeling on the transient segment data that we selected. This becomes the output data of the learning model. In step 3, for the classification model, we build the Seq2Seq LSTM model to train and test on generated features. The training process for HT-LSTM is illustrated in detail in Figure 2.

(2) Testing phase: In this phase, we use the trained Seq2Seq LSTM model to predict the generated testing data. As the training data process, we need to preprocess the raw testing data as the sequential testing data IA, IP, and IF. The testing phase is visualized in detail in Figure 3.

B. The APF Feature Generation Algorithm

In digital signal processing, we often use a discrete-time signal. We assume that a continuous signal denoted by $x(t)$ is sampled at intervals of T seconds, giving N real-valued discrete samples satisfying $x[n] = x(nT)$. Based on a discrete-time Fourier transform (DTFT), the spectrum X[f] can be constructed from the discrete-time signal $X[n]$, where f is the sampling frequency.

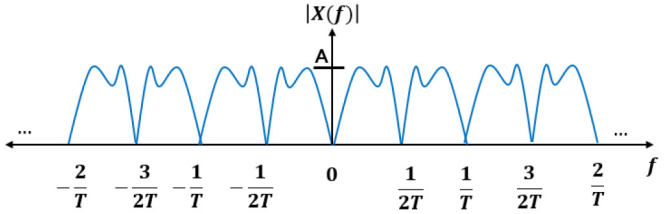


Fig. 4. The spectrum of a periodically sampled signal.

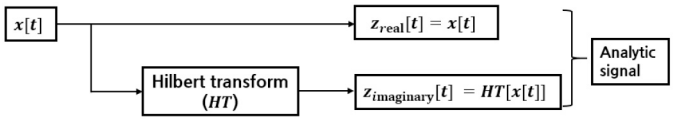


Fig. 5. The structure of the analytic signal.

$$X(f) = T \sum_{n=0}^{N-1} x[n] e^{-j2\pi fnT} \tag{1}$$

Figure 4 describes the spectrum of $x[n]$ resulted of periodically sampled signal $x[n] = x(nT)$.

In this sense, starting from a discrete real-valued sampled signal $x[n]$ we build $z[n]$, a discrete-time analytic signal, with $z[n]$ equal to $z_{real} + jz_{imaginary}[n]$, where j is the imaginary unit. Figure 5 depicts the structure of the analytic signal. In

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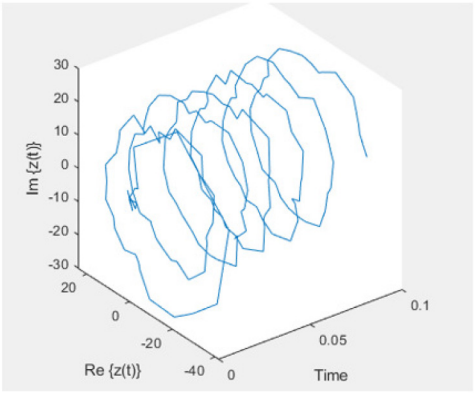


Fig. 6. An example of a complex helical sequence in 3D (An analytic signal example).

the continuous-time signal case, we can suppress the negative frequency components of the spectrum signal. However, for the discrete-time signal, we cannot perform this suppression it would remove the entire spectrum. One of the best solutions, in this case, is to remove the negative frequencies of each half spectral period by setting a zero value.

The above solution uses the HT algorithm, which was first introduced by Marple [28]. Suppose we have created a complex-valued N-point discrete-time signal. The HT process is as follows. First, compute $X[m]$ as the N-point DTFT based on FFT. Second, transform the one-sided analytic signal from the N-point discrete-time signal.

$$Z[m] = \begin{cases} X[0], & \text{if} \quad m = 0 \\ 2X[m], & \text{if} \quad 1 \le m \le \frac{N}{2} - 1 \\ X\left[\frac{N}{2}\right], & \text{if} \quad m = \frac{N}{2} \\ 0, & \text{if} \quad \frac{N}{2} + 1 \le m \le N - 1 \end{cases}$$

Third, compute the analytic signal $z[n]$ using an N-point inverse DFTF.

$$z[n] = \frac{1}{NT} \sum_{m=0}^{N-1} Z[m] e^{j2\pi mn/N} \tag{2}$$

In signal processing, IA, IP, and IF concepts are essential. HT returns a complex sequential helical sequence from real sequential data. The analytic signal is also called a helical sequence. Based on HT, we can calculate the instantaneous features of a time series. Specifically, these instantaneous features are the amplitude, phase, and frequency. IA is the amplitude of the complex HT. The average of frequencies in a time-varying signal is IF, given by the changing time pattern of the IP angle. IP reflects how the local phase angle varies linearly over a single cycle. In general, IA and IF are constant for a pure sinusoidal waveform. Figure 6 depicts one example of a complex helical sequence of Hilbert's result.

HT is notable in signal processing for providing a 90° phase shift compared to the input signal. For instance, if we choose the input signal to be a cosine function then calculating the HT will yield a negative sine function. While the frequency domain is transformed from the time domain by Fourier transform, HT keeps values in the time domain. The HT of a signal $x(t)$ is formed as a transform in which the phase angle of all components of the signals is shifted by ±90°. For any real-time series $x(t)$, the HT is calculated as follows:

$$HT(x(t)) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(u)}{t-u} du \tag{3}$$

With HT thus defined, the analytic signal including the time series and its transform can be formulated as follows:

$$S(t) = x(t) + jHT(x(t)) \tag{4}$$

where $HT(x(t))$ is the Hilbert transform (HT) of $x(t)$. The analytic signal can be presented in its exponential form as follows:

$$S(t) = A(t)e^{j\theta(t)} \tag{5}$$

where $A(t)$ is the IA and $\theta(t)$ is the IP.

The IF $f(t)$ or instantaneous pulse $w(t)$ can be defined as follows:

$$f(t) = \frac{w(t)}{2\pi} = \frac{1}{2\pi} \frac{d\theta(t)}{dt} \tag{6}$$

In the signal processing, HT can be computed in three steps: Step 1 is a calculation of the Fourier transform based on the *Forward_FFT* algorithm from the given signal $x(t)$; Step 2 is a rejection of the negative frequencies of the spectrum signal obtained, and Step 3 is a calculation of the inverse Fourier transform based on the *Inverse_FFT* algorithm. The results are then complex-value comprising a real part and an imaginary part. The forward FFT is used in HT. A divide and conquer strategy is used to solve the FFT problem. For faster multiplication of polynomials, FFT uses complex roots of unity. A complex n^{th} root of unity is a complex number Ω satisfying $\Omega^n = 1$. The inverse FFT is also used in the HT process. We can compute inverse FFT in time $O(NlogN)$ in three steps. Step 1 is switching the roles of x and X. Step 2 is replacing Ω_N by Ω_N^{-1}, and Step 3 is dividing each element of the result by N. Thus, the analytic signal is created by Algorithm 1.

In this section, we describe the pseudocode of the procedure for generating the APF feature. APF is created by extracting IA, IP, and IF values. The process of HT-LSTM is shown in Algorithm 2, whose output is APF. APF is used in turn as the input for the Seq2Seq LSTM model.

C. Applying the Seq2Seq LSTM Approach to APF Data

Long Short-Term Memory (LSTM) is an extension of Recurrent Neural Network (RNN) first published in 1997 by Hochreiter and Schmidhuber [29]. The LSTM cell is designed as a unit of logic, and as such helps to reduce the vanishing gradient problem significantly. It makes the RNN model more useful for long-term memory storage as an internal memory state.

The Seq2Seq LSTM model is suitable for classification problems in sequential data. In this study, we use this model because the training and testing processes have sequential input *APF* denoted by x and output data $L^{'}$ denoted by y as notations in Figure 7. This figure depicts the structure of the Seq2Seq LSTM model. The Seq2Seq LSTM model includes four main layers: an input gate, a hidden layer, a softmax layer, and an output gate. The hidden layer contains the LSTM cells.

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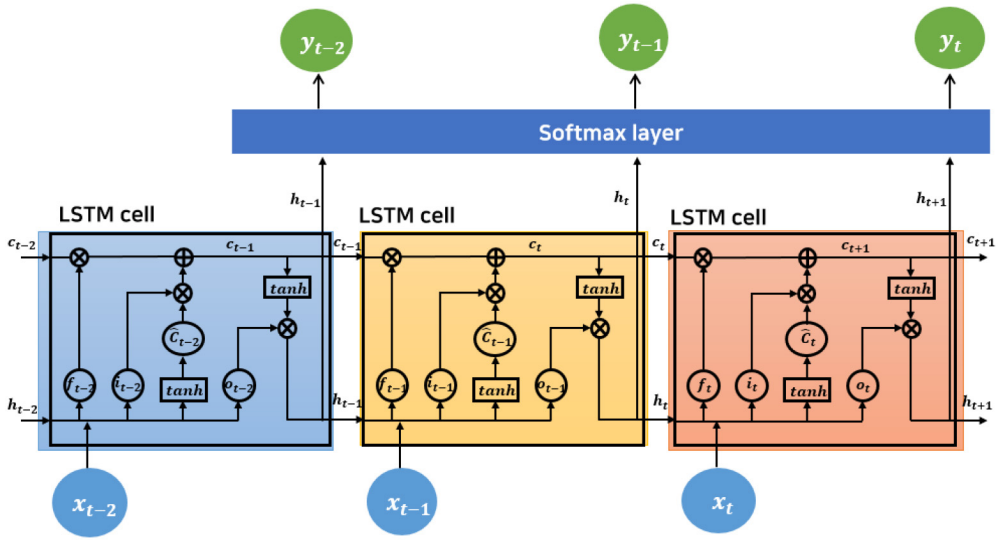


Fig. 7. The architecture of Seq2Seq LSTM model.

Algorithm 1 HT

Require: Original real-value signal $x[n]$

Ensure: Analytic signal $z[n]$

1: $N \leftarrow len(x[n])$

Transform the N-point discrete time Fourier from $x[n]$ based on Forward_FFT algorithm:

2: $X[n] \leftarrow Forward_FFT(x[n])$

Create the one-side analytic signal by rejecting negative frequency as the following equation:

3: $Z[m] \leftarrow X[0], \quad if \quad m = 0$

$Z[m] \leftarrow 2X[m], \quad if \quad 1 \le m \le \frac{N}{2}$

$Z[m] \leftarrow X[N/2], \quad if \quad m = \frac{N}{2}$

$Z[m] \leftarrow 0, \quad if \quad \frac{N}{2} + 1 \le m \le N - 1$

Based on Inverse_FFT algorithm, the analytic signal can be achieved:

4: $z[n] \leftarrow Inverse_FFT(Z[m])$

5: **return** $z[n]$

Algorithm 2 Create *APF* Sequential Data Are Transient-State Data

Require: I is raw current data, Fs is sampling frequency

Ensure: *APF*

1: Identify the RMS threshold th which occurred event of I

2: Identify transient segment data of I: $Seg_I \leftarrow I[th - Fs : th + Fs]$

3: Create analytic signal(AS): $AS \leftarrow \mathbf{HT}(Seg_I)$, where HT is Hilbert transform

4: $IA \leftarrow sqrt(real^2(AS) + imaginary^2(AS))$

5: $IP \leftarrow arctan\left(\frac{imaginary(AS)}{real(AS)}\right)$

6: $IF \leftarrow \frac{IP}{2\pi \times Fs}$

7: Create matrix data APF: $APF \leftarrow [IA, IP, IF]$

8: Create sequential data from matrix data by transpose matrix operation: $APF = APF^T$

9: **return** *APF*

The softmax layer is added because the output layer contains categorical and sequence data.

In particular, the three gates in each LSTM cell are the input gate i_t, the forget gate f_t, and the output gate o_t, along with the cell state c_t. A key feature of the Seq2Seq LSTM model that it is many to many with time by time. Moreover, the number of input nodes is the same as the number of outputs in this model. We take $(x_t^1, x_t^2, x_t^3, \ldots, x_t^n)$ as a sequence input vector and $(h_t^1, h_t^2, h_t^3, \ldots, h_t^n)$, $(y_t^1, y_t^2, y_t^3, \ldots, y_t^n)$, $(c_t^1, c_t^2, c_t^3, \ldots, c_t^{n_c})$ as hidden, sequence output and cell vectors, respectively. The elements of each vector are units for the layers of an LSTM RNN. The number of units in each layer is n, and the weight matrix superscripts have the obvious meaning. For example, W_t^{hi} is the hidden-input gate weight matrix, and W_t^{xo} is the input-output gate weight matrix. Bias terms are denoted b_i, b_f, b_t, and b_o .

First, we compute the value of f_t, the activation of the forget gate. The range value of the output f_t is from 0 to 1 and this output value will be multiplied by c_{t-1} when calculating c_t. Therefore, f_t represents the activation rate of the previous cell state:

$$f_t = sigmoid\left(W_t^{xf} \sum_{i=1}^{n} x_t^i + W_t^{hf} \sum_{j=1}^{n} h_{t-1}^j + b_f\right) \tag{7}$$

Second, we calculate the value for the input gate i_t. As with f_t, i_t is the activation ratio of the input value x_t.

$$i_t = sigmoid\left(W_t^{xi} \sum_{i=1}^{n} x_t^i + W_t^{hi} \sum_{j=1}^{n} h_{t-1}^j + b_i\right) \tag{8}$$

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Third, we measure the value for the cell state c_t, combining two factors: the forget gate and the input value activated by the input gate, where notations $\otimes$ and $\oplus$ are denoted by element-wise multiplication and element-wise addition operations, respectively.

$$\hat{C} = tanh\left(W_t^{xc}\sum_{i=1}^{n}x_t^i + W_t^{hc}\sum_{j=1}^{n}h_{t-1}^j + b_c\right) \tag{9}$$

$$c_t = f_t \otimes c_{t-1} \oplus i_t \otimes \hat{C} \tag{10}$$

Finally, we obtain the output gate value and use it for the memory block output.

$$o_t = sigmoid\left(W_t^{xo}\sum_{i=1}^{n}x_t^i + W_t^{ho}\sum_{j=1}^{n}h_{t-1}^j + b_o\right) \tag{11}$$

$$h_t = o_t \otimes tanh(c_t) \tag{12}$$

Equations (7) to (12) are processed in one LSTM cell. After all processing in the hidden layer is complete, we can calculate the output units with the hidden vector h_t.

$$y_t = softmax\left(W_t^{hy}\sum_{i=1}^{n}h_t^i\right) \tag{13}$$

The goal of Seq2Seq LSTM is the probability vector $p(y_1, \ldots, y_t|x_1, \ldots, x_t)$. The conditional probabilities are computed based on first obtaining the fixed-dimensional representation v of the sequential input, given by the last hidden state in the LSTM model. The probabilities for the sequential output are then computed with the following formulation:

$$p(y_1, \ldots, y_t|x_1, \ldots, x_t) = \prod_{i=1}^{t} p(y_t|v, y_1, \ldots, y_{t-1}) \tag{14}$$

where each distribution $p(y_t|v, y_1, \ldots, y_{t-1})$ passes over all the labels in the output sequence via the softmax function.

The deep neural network (DNN) optimization method is critical to improving the accuracy of the inference model. Several optimization techniques have shown good results for DNNs. This section details the necessary and efficient methods used for DNN models, especially LSTM, and Seq2Seq LSTM models. The basic optimization method is stochastic gradient descent (SGD), with the Adam method as an efficient choice. LSTM models have used a back-propagation algorithm to update the weights. In this algorithm, the gradient must be computed many times to adjust the weights of the neural networks. Computing the gradient will become very impractical if a training set is too big so that the model is trained too slowly to fit in memory. To solve this problem, SGD is used. In this method, the gradient is computed over a few examples instead of all training sets. The advantage of SGD is the ability to create the local minimal [30] via a mini-batch. Generally, mini-batches are often used because it is possible to reduce learning variance, giving the model more stable convergence. Another reason is that with the high computational power of GPUs, mini-batches can be processed very fast since the operation is easily parallelized. To avoid problems with SGD, Adam is one of the best solutions. Adam optimizer is calculated following equation (15). In addition, a first-order gradient-based method introduced by Kingma and Ba [31], is a popular algorithm as an optimizer for deep learning models because it achieves good results as fast as possible in large models and datasets. In summary, Adam is a good choice because it is fast and does not need too strict a learning schedule.

TABLE I
HYPER-PARAMETER SETTING FOR SEQ2SEQ LSTM IN HT-LSTM METHOD

Hyper-parameter	Value
Optimizer	Adam
Number of hidden layers (LSTM layers)	1
Number of features	3
Number of hidden nodes	200
Mini batch-size	100
Number of epochs	1000
Learning rate	0.001
Iteration	50

$$\begin{aligned} g_t &= \nabla_\theta f_f(\theta_{t-1}) \\ m_t &= \beta_1 * m_{t-1} + (1-\beta_1) * g_t \\ v_t &= \beta_2 * v_{t-1} + (1-\beta_1) * g_t^2 \\ \hat{m}_t &= \frac{m_t}{1-\beta_1^t} \\ \hat{v}_t &= \frac{v_t}{1-\beta_2^t} \\ \theta_t &= \theta_{t-1} - \alpha * \frac{\hat{m}_t}{\sqrt{\hat{v}_t}+\epsilon} \end{aligned} \tag{15}$$

where α is the step size or learning rate; β_1, $\beta_2 \in [0, 1]$ are exponential decay rates for the moment estimates; ϵ is a tiny number to prevent any division by zero in the implementation. The initial conditions are $m_0 = 0$ and $v_0 = 0$. The good default settings for the setting machine learning parameters are $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$.

In HT-LSTM, the Seq2Seq LSTM's hyper-parameter setting is significant for obtaining the best possible accuracy of this learning model. Hyper-parameter values for the Seq2Seq LSTM model learning on APF features are shown in Table I. APF feature contains three data sequences IA, IP, and IF as for the input learning of the LSTM model. Therefore, the number of features is 3. Because the number of hidden nodes and mini-batch size affect Adam optimizer as well as training time of the learning LSTM model. Hence, while doing our experiments, we assigned the number of hidden nodes and mini batch-size values are 200 and 100, respectively. These selected values guarantee our LSTM model trained is not too slowly or too quickly on the APF feature to obtain more stability and convergence in the training process. Via the experiment, we run our model on more than two LSTM hidden layers, however, the training process is not stable and convergence of accuracy and loss at several epochs. Hence, we found the number of hidden layers equals 1 is the best value for our learning method.

IV. EXPERIMENT AND DISCUSSION

We perform experiments on high-frequency NILM public datasets including the BLUED (Building-Level fUlly-labeled dataset for Electricity Disaggregation) and PLAID (Plug Load

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TABLE II
APPLIANCES SELECTED AND THEIR LOAD TYPES IN BLUED DATASET

Appliance	Type	Phase Label
Fridge	Inductive load	A1
Light1 (Backyard light)	Resistive load	A2
High-Power1 (Kitchen aid chopper)	Inductive load	A3
Light2 (Basement light)	Resistive load	B1
High-Power2 (Iron)	Resistive load	B2
Computer and monitor (LCD monitor)	SMPS load	B3

Appliance Identification Dataset) dataset in our method evaluation as well as a comparison with other existing methods on these datasets. This selection because BLUED and PLAID datasets are provided available at high sampling rate data which are several prior NILM methods used for these datasets on their applications. Hence, we could compare the proposed method to their works. Furthermore, most NILM research focuses on this low-frequency dataset while existing NILM research on the high-frequency dataset is rare. Although REDD and UK-DALE datasets are frequently used in NILM prior researches, however, most researchers used these datasets at low-frequency data. Hence, we used BLUED and PLAID datasets at a high-frequency data for our experiment on evaluation and comparison in this work.

The publicly available BLUED dataset is used for evaluating the proposed method in this study for the following reasons. This dataset consists of labeled data for single household appliances. The dataset was collected in two phases, A and B, with each phase containing several appliances. These appliances have various load types, for example, inductive, resistive, and SMPS load. Also, prior research has used this data for performing experiments on the transient-state approach, corresponding closely with our proposed approach. Hence, this dataset is suitable for our proposed approach. The BLUED dataset [32] was collected in only 8 days and has dozens of appliances. Hence, each appliance has a very small sample data size. For comparison with the Fast Shapelets [33] and STFT with CNN [14], in each appliance group, we chose the representative linear or non-linear loads, for example, inductive load, resistive load, or SMPS load. Table III shows the type information for each load. The dataset is chosen to evaluate the proposed method for detecting various types of load. In data preprocessing, we down-sampled the BLUED dataset at a 12 kHz to 2 kHz sampling rate.

Another high-frequency NILM data, PLAID, in which is used to perform our experiment evaluation. PLAID [34] dataset comprises short voltage and current measurements at 30 kHz from 11 different appliances in houses in Pittsburgh, Pennsylvania, USA. Therefore, we select these appliances in houses randomly and category each load type, which is shown in Table III. In doing the experiment, we also down-sampled the data to a 2 kHz sampling rate.

Besides using the public NILM data at high frequency for evaluation, we use a proprietorial private dataset to evaluate the proposed method. In this private dataset, collected at the high-frequency by 15 kHz sampling, household appliance data was collected in the lab environment. Five individual home appliance data case and four mixed-load data case were collected

TABLE III
ELEVEN APPLIANCES SELECTED AND THEIR LOAD TYPES IN THE PLAID DATASET

Appliance	House	Type
Incandescent light bulb	1	Resistive load
Compact fluorescent lamp	5	Resistive load
Fridge	9	Inductive load
Microwave	1	Resistive load
Washing machine	6	Inductive load
Air conditioner	2	Resistive load
Fan	2	Inductive load
Hairdryer	2	Resistive load
Heater	2	Resistive load
Laptop	2	SMPS load
Vacuum	6	Inductive load

TABLE IV
SINGLE LOAD TYPES AND MIXED LOADS ON THE PRIVATE DATASET

Appliance	Type
Fan	Inductive load
Hairdryer	Resistive load
Kettle	Resistive load
LG monitor	SMPS load
Samsung monitor	SMPS load
Samsung monitor-Kettle	Mixed load
Fan-Samsung monitor	Mixed load
Fan-Kettle	Mixed load
Samsung monitor-Kettle-Fan	Mixed load

TABLE V
PERFORMANCE DETECTION OF HT-LSTM ON BLUED DATASET

Appliance	Precision	Recall	Accuracy	F1
Fridge	0.9456	0.9305	0.8832	0.938
Light1	0.8812	0.946	0.839	0.9125
High-Power1	0.9674	0.9335	0.905	0.9501
Light2	0.9444	0.9535	0.9015	0.9482
High-Power2	0.9105	0.9305	0.8525	0.9204
Computer and monitor	0.9852	0.9335	0.9206	0.9586
Average	0.9391	0.9377	0.8836	0.938

(see Figure 4). Similar to the two public datasets above, we also down-sampled this data to a 2 kHz sampling rate in our experiment.

For performance evaluation of the proposed method, we use classification performance measures including accuracy, precision, recall, F1, following the equations that were mentioned in [35]. We perform our proposed method on the computational environment as following: Intel Core i7-4790 CPU @ 3.60GHz; RAM:16GB; Windows 10 Professional 64-Bit, MATLAB R2020a.

A. Evaluation HT-LSTM

The classification results for the Seq2Seq LSTM model on two public datasets, BLUED and PLAID, and a private dataset are shown in this section. We give a detailed breakdown of these results below.

- Classification results on BLUED dataset: The average accuracy and F1 on the BLUED dataset were 88.36% and 93.8%, respectively. Table V shows the performance of HT-LSTM on the BLUED dataset in detail. In this dataset, three load types are categorized which comprise inductive load (Fridge and Kitchen aid chopper), resistive load

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TABLE VI
PERFORMANCE DETECTION OF HT-LSTM ON PLAID DATASET

Appliance	Precision	Recall	Accuracy	F1
Incandescent light bulb	0.9271	0.98	0.9099	0.9528
Compact fluorescent lamp	0.9181	0.976	0.8978	0.9461
Fridge	0.9746	0.959	0.9355	0.9667
Microwave	0.9267	0.987	0.9155	0.9559
Washing machine	0.9309	0.9439	0.8821	0.9374
Air conditioner	0.9861	0.9239	0.9121	0.954
Fan	0.9789	0.9309	0.9127	0.9543
Hair-dryer	0.9785	0.9129	0.8945	0.9446
Heater	0.9758	0.97	0.9472	0.9729
Laptop	0.9218	0.8849	0.8231	0.903
Vacuum	0.9356	0.9749	0.9136	0.9549
Average	0.9504	0.9494	0.904	0.9493

(Backyard light and Basement light, and Iron), and SMPS load (LCD monitor). While accuracy and F1's average measurements of resistive load are 86.43% and 92.7%, respectively, these accuracy and F1 measurements are better on inductive load by 89.41% and 94.41%, respectively; and then these measurements are the best accuracy and F1 results of 92.06% and 95.86% on SMPS load, respectively. The performance classification on the resistive load of the BLUED data selected is not better than others because Backyard light and Basement light belong to several kinds of light devices.

- Classification results on the PLAID dataset: The average accuracy of the proposed method on the PLAID dataset was 90.04%. The average F1 measurement of the proposed method on the dataset was 94.93%. Table VI depicts the performance of HT-LSTM on the PLAID dataset in detail. In the PLAID dataset, three load types consist of inductive load (Fridge, Washing machine, Fan, and Vacuum), resistive load (Incandescent light bulb, Compact fluorescent lamp, Microwave, Air conditioner, Hairdryer, and Laptop), and SMPS load (Laptop). The performance classification results of SMPS load are 83.31% average accuracy and 95.49% average F1. The higher result of average accuracy and F1 measurement on inductive load (91.1% for accuracy and 95.33% for F1) and resistive load (91.29% for resistive load and 95.44% for F1) compare to SMPS load. Although the average accuracy of SMPS load is lower than inductive load and resistive load, F1 measurement of SMPS load, resistive load, and inductive load are high with over 95%.
- Classification results on the private dataset: The average accuracy and F1 on the private dataset were 93.4% and 94.23%, respectively. Table VII shows the performance of HT-LSTM on the private dataset in detail. Each load type contains each load is shown in Table IV. The average results of accuracy and F1 are high-performance classification on inductive load and resistive load. In particular, the inductive load has 93.12% for accuracy and 96.44% for the F1 result. Besides, the resistive load has 98.91% for accuracy and 99.45% for F1 measurement. However, the SMPS has lower accuracy and F1 measurements with 79.43% for accuracy and 88.3% for F1 result. This because the SMPS load contains similar monitor load devices, but different brands, Samsung and LG.

TABLE VII
PERFORMANCE DETECTION OF HT-LSTM ON THE PRIVATE DATASET

Appliance	Precision	Recall	Accuracy	F1
Fan	0.9814	0.948	0.9312	0.9644
Hairdryer	0.993	0.998	0.9911	0.9955
Kettle	0.9891	0.998	0.9871	0.996
LG monitor	0.9599	0.9109	0.8775	0.9935
Samsung monitor	0.7244	0.975	0.7112	0.8312
SM-Kettle	0.999	0.994	0.993	0.9965
Fan-Samsung monitor	0.9408	0.715	0.6842	0.8125
Fan-Kettle	0.995	0.988	0.9831	0.9915
Samsung monitor-Kettle-Fan	0.9938	0.9708	0.965	0.9822
Average	0.9529	0.9422	0.9026	0.9447

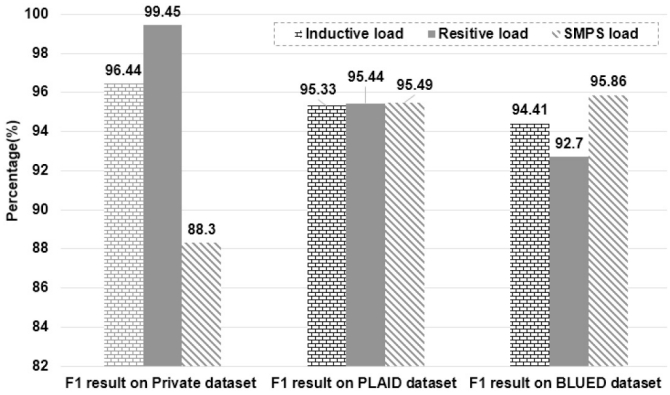


Fig. 8. F1 measurement result of HT-LSTM for each load type on each dataset.

TABLE VIII
COMPARISON OF CLASSIFICATION PERFORMANCE OF HT-LSTM WITH OTHER METHODS ON BLUED DATASET

Method	Recall	Precision	Accuracy	F1
Fast Shapelets [33]	77.6	69.7	-	72.3
Cross-dataset evaluation + SVM [36]	-	-	80	-
Concatenate-DenseNet-121 [14]	80.7	90.0	-	83.9
HT-LSTM	**93.91**	**93.77**	**88.36**	**93.8**

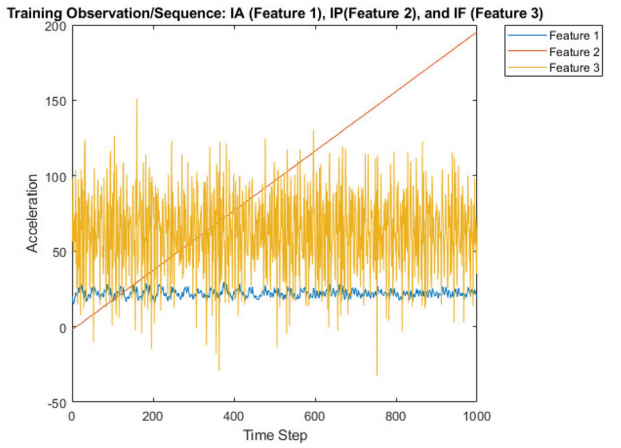
In addition, we category the performance classification F1 measurement of the HT-LSTM for each three load type (inductive load, resistive load, and SMPS load) of three datasets (private dataset, PLAID dataset, and BLUED dataset with high sampling rates). Figure 8 is summarized from the results of three Tables V, VI, and VII.

B. Comparing Classification Performance to Previous Methods

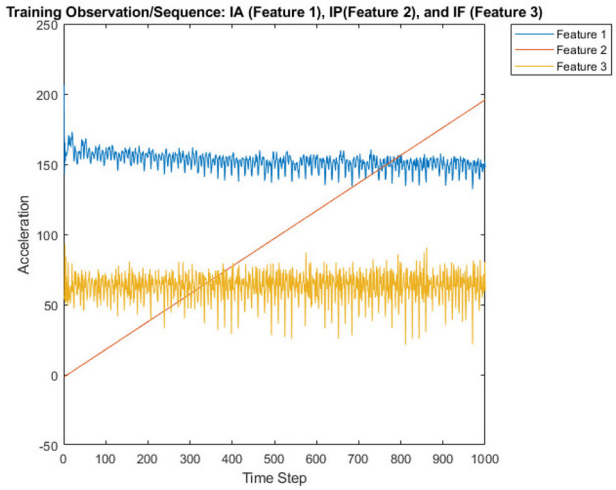
In this section, we compare the recognition performance of HT-LSTM to other methods on two similar publicly available datasets, BLUED and PLAID.

- To BLUED dataset, we find that the proposed HT-LSTM outperforms prior methods in both accuracy and F1 measurements. Looking in particular at the prior method with the highest classification performance, Concatenate-DenseNet-121, the proposed HT-LSTM method shows a performance increase of 9.9% for the F1 measurement. Detailed comparison information is presented in Table VIII.

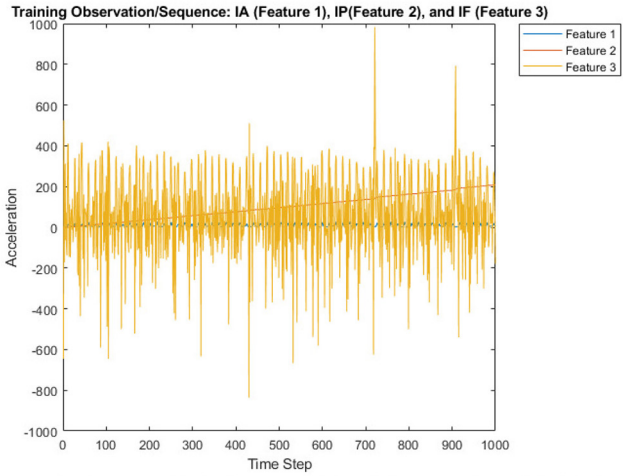
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(a) Sequences IA, IP, IF generated from Fan data (Inductive load)



(b) Sequences IA, IP, IF generated from Hair-dryer data (Resistive load)



(c) Sequences IA, IP, IF generated from LG monitor data (SMPS load)

Fig. 9. IA, IP, IF sequential patterns on each load type.

- To PLAID dataset, although the accuracy of the proposed method is less accurate with the highest accuracy of FFT-BDT about 2.76%, our method presents outperform classification performance F1 measurement of 2.45%. Table IX shows the performance comparison of the proposed method with others on the same dataset, PLAID.

TABLE IX
COMPARISON OF CLASSIFICATION PERFORMANCE OF HT-LSTM WITH OTHER METHODS ON PLAID DATASET

Method	Accuracy (%)	F1(%)
VI based on EFDs [37]	79	-
VI image-CNN [38]	-	87.95
DWT-Decision Tree [22]	90	-
Harmonic current-MLP [39]	75.38	-
HAR (Harmonic) [40]	68.9	67.5
VI tracjectories-Neural network ensemble [25]	89.7	-
CW (Current Waveform)[40]	90.9	89.8
VI trajectories-Siamese ANNs+DBSCAN [41]	-	90
kNN (k-nearest neighbor) [42]	-	90
FFT-BDT [15]	**92.8**	92.48
HT-LSTM	90.04	**94.93**

In summary, the proposed method outperforms previous methods about the classification performance of household appliances are performed to the same high-frequency public datasets, BLUED and PLAID.

C. Discussion

Our novel APF feature can support overcoming the current NILM challenge because its component features are generated from three kinds of load (inductive, resistive, and SMPS) with different and unique features. Each load type has different IA, IP, IF signals generated by analytic signals by applying the Hilbert transform procedure. Moreover, the APF features are obtained on differences in transient time and shape for these load types. In summary, the APF feature has two characteristics. The first is ease of distinguishing between APF features because the pattern of each load type is different and unique. The second is the obtained APF features which are generated from differences in the transient time and shape of loads, in particular the three types of loads, namely inductive (e.g., fan), resistive (e.g., hairdryer) and SMPS load, e.g., (LG monitor). We depict the APF features obtained from three load examples corresponding to three load types in Figures 9(a), 9(b), and 9(c).

In addition, the results of Algorithm 2 form the input for the learning model, Seq2Seq LSTM. In this study, we adopt Seq2Seq LSTM to learn about APF features for load classification for the following reasons. First, APF features are extracted based on HT. Since the HT algorithm is based on the FFT algorithm, it likewise has fast computation. APF features are time-series data. Hence, they constitute sequential data. Second, although the RNN model is suitable for learning sequence data, it suffers from a gradient vanishing problem. The LSTM model is one of the best solutions for overcoming this problem with RNN. Third, in a real-time NILM system, home appliances can be used at the same time. This means that many different events can occur, for example, the fan and the hairdryer operating at the same time. Therefore, we use a labeling sequence is to present these varied events. For this reason, it is advantageous to apply Seq2Seq LSTM to learn on the APF feature to recognize loads.

Although many researchers proposed their NILM methods to provide more accurate load identification or load disaggregation, almost their NILM results are focused on low-frequency public NILM datasets, for example, UK-DALE,

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REDD, and so on, and even the private datasets from their data acquisition. The numerical results are focused on the classification to solve the NILM issue using high-frequency are so rare. Hence, in this work, we investigate a novel approach to build a load identification model on high-frequency data to improve load classification. Through our performance results along with the performance comparisons to other prior methods on two public datasets, BLUED and PLAID, the NILM method used high-frequency data is promising more attraction and investment in research and development to solve the NILM issues in future work.

On the other hand, almost numerical results are focused on the classification to solve the NILM issue which includes load identification and load disaggregation. Nevertheless, some other researches are focused to forecast future energy consumption by regression solutions based on deep learning, for examples, LSTM recurrent neural network-based framework [43], probabilistic load forecasting based on Bayesian deep learning [44], the proposed deep mixture architecture between CNN (Convolutional Neural Network) and GRU (Gated Recurrent Unit) [45] in recently. The regression which finds future actual consumption patterns is a promising difference issue in future power consumption management.

V. Conclusion

In this work, we propose a novel method, HT-LSTM, for improving performance at the load identification task, solving a key current challenge in NILM using high-frequency data. HT-LSTM consists of two main components: the generation of the novel feature group APF, comprising sequences IA, IP, and IF based on HT algorithm; and the Seq2Seq LSTM model used for load classification. The obtained APF sequence is a unique feature for each appliance. Hence, this feature supports the LSTM model enhances its performance classification in recognition of residential load types including inductive load, resistive load, and SMPS load. We conduct experiments to evaluate our proposed method on two public datasets, BLUED and PLAID, and a private dataset collected at the lab. By overall performance measurement, F1 results show that our model outperforms previous models on similar high-frequency public datasets. Through the experimental results obtained, we find that our proposed method can recognize various load types (inductive load, resistive load, and SMPS load) that contain transient shapes for various household appliances. Almost numerical results are focused on the classification to solve the NILM issue using low-frequency data, but a few researchers focus their method on NILM using high-frequency data. By our experimental results on two high-frequency public datasets, the opportunity for the improvement and development of NILM methods using high-frequency data is left open and promises to attract more attention in the future. Furthermore, in future works, we focus on regression issues to research and develop a forecasting model for future electric energy usage.

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